Gondwana Research 41 (2017) 337–351

Contents lists available at ScienceDirect

Gondwana Research

journal homepage: www.elsevier.com/locate/gr

New Late Cretaceous paleomagnetic data from volcanic rocks and red beds from the Lhasa terrane and its implications for the paleolatitude of the southern margin of Asia prior to the collision with India

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ARTICLE INFO

Article history:
Received 31 May 2015
Received in revised form 17 October 2015
Accepted 3 November 2015
Available online 23 December 2015

Keywords:
Lhasa terrane
Late Cretaceous
Paleomagnetism
India–Asia collision
Crustal shortening

ABSTRACT

In order to improve estimation of the paleolatitude of the southern margin of Asia during the Late Cretaceous, as well as the constraints on the intracontinental convergence within Asia due to the India–Asia collision, we performed a combined geochronological and paleomagnetic investigation of Late Cretaceous red bed sequences and intercalated volcanic rocks from the Linzhou Basin in the Lhasa terrane (29.9°N/91.1°E). The results of potassium–argon (K–Ar) dating indicated that the basalt sequences are of Late Cretaceous age (68–75 Ma). Stepwise thermal demagnetization successfully isolated high unblocking-temperature characteristic directions from 10 red bed sites and 11 lava flow sites. Positive fold and reversal tests indicate a primary origin for the characteristic remanence of the red beds and basalt sites. The consistent inclination recorded in the red beds and basalt lava flows indicates that no significant compaction-related inclination shallowing has occurred in the red beds from the studied section. The tilt-corrected mean direction is D/I = 0.5°/20.2° with α_{95} = 6.4° and N = 21 sites, corresponding to a paleopole at 70.5°N, 269.6°E with A_{95} = 4.9°. The paleomagnetic results yield a paleolatitude of 10.4 ± 4.9°N for the southern margin of Asia in the Late Cretaceous. Compared with expected paleomagnetic directions from the Tethyan Himalaya and Asia, significant crustal shortening of 17.7 ± 5.5° (1960 ± 610 km) and 9.4 ± 4.7° (1040 ± 520 km), respectively, may have occurred between the southern margin of Asia and the Tethyan Himalaya, and within Asia, since the Late Cretaceous. Based on previous Cretaceous and Paleocene data from the Lhasa terrane, the latitude of the southern margin of Asia was located at a stable paleolatitudinal position (10°–15°N) during the Cretaceous and Paleocene.

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1. Introduction

The collision of the Indian and Asian continents is considered to be one of the major tectonic events of the Cenozoic. It resulted in the formation of the Himalayas and the Tibetan Plateau (Yin and Harrison, 2000; Ding et al., 2005; Zhang et al., 2014, 2017), which may have had major repercussions for regional environmental, and possibly global climatic change (Tapponnier et al., 2001; Royden et al., 2008; Boos and Kuang, 2010; Molnar et al., 2010). Despite its importance and several decades of research, the precise timing of the initial India–Asia contact and the amount of crustal shortening have remained controversial (DeCelles et al., 2002; Aitchison et al., 2007; Garzanti, 2008). The proposed initial timing of the collision mainly ranges from ~34 Ma to ~70 Ma and is based mainly on the following evidence: A dramatic decrease in the rate of motion of the Indian plate relative to Asia (Patriat and Achache, 1984; Klootwijk et al., 1994); the end of marine sedimentation in the Tethyan Sequence, and the appearance of suture-zone and Himalayan foreland basin strata (Gaetani and Garzanti, 1991; Willems et al., 1996; Cai et al., 2011); biological connectivity between the Indian and Asian continents (Rage et al., 1995); the syn-collision or post-collision volcanic rocks (Ding et al., 2005; Mo et al., 2007); paleomagnetic results (Besse et al., 1984; Tong et al., 2008; Dupont-Nivet et al., 2010a,b; Liebke et al., 2010; Tan et al., 2010; Yi et al., 2011; Meng et al., 2012; Sun et al., 2012; Chen et al., 2014; Ma et al., 2014; Yang et al., 2015) and combinations of the above evidence (Yin and Harrison, 2000; Aitchison et al., 2007; Najman et al., 2010; White and

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http://dx.doi.org/10.1016/j.gr.2015.11.006
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Lister, 2012). Aitchison et al. (2007) proposed a very late age (~34 Ma) for the initial India–Asia collision, implying that the collision should have occurred at the present position of the Yarlung Zangbo Suture Zone (YZBS) and that no apparent post-collisional crustal shorting has occurred within Asia. These hypotheses are all in conflict with recent paleomagnetic results from the Lhasa terrane (Chen et al., 2010; Liebke et al., 2010; Sun et al., 2010, 2012; Tang et al., 2013; Ma et al., 2014; Yang et al., 2015), which suggest a more southerly position (~10–16°N) of the Lhasa terrane during the Cretaceous and Paleocene.

Most of the existing paleomagnetic data from the Tibetan terrane suggest that some 2600–3400 km of post-collision convergence has taken place between Asia and greater India, with 1100–2000 km of this total length accommodated by north–south shortening in the Tibetan plateau and lateral tectonic escape (Achache et al., 1984; Patriat and Achache, 1984; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Sun et al., 2010; van Hinsbergen et al., 2012; Tang et al., 2013; Ma et al., 2014; Li et al., 2017; Yi et al., 2015). However, Tan et al. (2010) propose a much smaller degree of continental shortening for the Late Cretaceous lava flows (200–600 km) and for the Eocene tuffs (0 km) within Tibet. However, this is in conflict with geological observations (Yin and Harrison, 2000; DeCelles et al., 2002; Johnson, 2002; Guillot et al., 2003) and global deeper seismic tomography (Replumaz et al., 2013).

Knowledge of the pre-collisional paleolatitudinal positions of the continental margins of southern Eurasia (the Lhasa terrane) is crucial for determining the initial age and the amount of crustal shortening between India and Asia. Paleomagnetism is the only effective method for directly determining paleolatitudes. Previous paleomagnetic studies of the Cretaceous–Cenozoic sediments (or volcanic rocks) in the Lhasa terrane have played a key role in paleoposition reconstructions of the southern margin of Asia (Westphal et al., 1983; Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993; Sun et al., 2008, 2012; Chen et al., 2010; Tan et al., 2010; Liebke et al., 2010; Dupont-Nivet et al., 2010a,b; Chen et al., 2012; Meng et al., 2012; van Hinsbergen et al., 2012; Huang et al., 2013; Tang et al., 2013; Chen et al., 2014 Ding et al., 2014; Ma et al., 2014; Huang et al., 2015a,b; Li et al., 2017; Yang et al., 2015; Yi et al., 2015); however, there are large uncertainties in these paleomagnetic data. Several paleomagnetic results from Cretaceous and Cenozoic volcanic rocks imply a low paleolatitude of ~10°N to ~15°N for the Lhasa area (Achache et al., 1984; Otofuji et al., 1991; Sun et al., 2008, 2010; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Chen et al., 2012; Tang et al., 2013; Ma et al., 2014; Li et al., 2017; Yang et al., 2015; Yi et al., 2015). However, several studies have reported a more northerly paleolatitude of 24°–32° N for the Lhasa area (Tan et al., 2010; Dupont-Nivet et al., 2010a; Meng et al., 2012; Huang et al., 2013, 2015a).

The foregoing review demonstrates that the existing Late Cretaceous and Cenozoic paleomagnetic results from the Lhasa terrane are limited and controversial. The disparities between published paleomagnetic data sets may be attributed to low-latitude bias due to paleomagnetic inclination shallowing in sediments during depositional and lithification processes (Tan et al., 2010; Huang et al., 2013; Ding et al., 2014; Huang et al., 2015a), together with remagnetization and secular variation of the Earth's magnetic field (Huang et al., 2015b). However, additional paleomagnetic studies are needed to resolve this issue.

In the present paper, we present paleomagnetic results from Late Cretaceous red beds sequences intercalated with basalt flows in the Linzhou Basin in the Lhasa terrane in order to provide reliable paleomagnetic constraints on the paleogeography of the southern edge of the Asian continent during the Late Cretaceous.

2. Geological setting and sampling

The Tibetan plateau is located between India to the south and the Tarim terrane to the north (Chang and Cheng, 1973). It can be divided primarily into five tectonic units, from south to north: the Himalayan, Lhasa, Qiangtang, Songpan–Ganze and Qaidam terranes (Fig. 1). These terranes are separated by several sutures: the India and Lhasa terranes are separated by the Yarlung–Zangbo suture, the Bangong–Nujiang suture separates the Lhasa terrane from the Qiangtang terrane, the Qiangtang terrane is separated from the Songpan–Ganze terrane by the Jinsha suture, and the Ayimaqin–Kunlun–Muztagh suture separates the Songpan–Ganze and Qaidam terranes (Yin and Harrison, 2000; Kapp et al., 2007). The Lhasa terrane was accreted to the southern margin of Eurasia prior to the Cenozoic India–Asia collision (Tapponnier et al., 2001). It extends more than 2000 km from east to west and about 300 km from north to south. The Cretaceous strata in the Lhasa terrane are more than 3 km thick and the Upper Cretaceous strata consist of non-marine fluvial red beds (He et al., 2007).

The Shexing Formation exposed in the Lhasa terrane overlies the Early Cretaceous Takena Formation and is unconformably overlain by the Paleogene volcanic strata of the Linzizong Group. In the study area, the Shexing Formation consists of red beds interbedded with basalt lavas. Fossils identified in the formation include Ostracoda (*Altanicypris*, *Cypridopsis*) and pollen and spores (*Retitricolpites georgensis*, *Tricolpites*, *Cyathidites minor*, *Cyathidites australis*, *Densoisporites*, *Classopollis* sp., *Cerebropollenites* sp., *Biretisporites* sp.), indicative of a Late Cretaceous age (BGMRXAR, 1993). Fossils in the Shexing Formation are unsuitable for unambiguous biostratigraphic age determinations; however, the ages of the underlying Takena Formation and overlying Linzizong Group are well constrained. The youngest fossils in the Takena Formation are Albian (BGMRXAR, 1993). On the basis of detailed mapping and eruption hiatuses, the Linzizong Group volcanic sequences can be subdivided into the Lower Dianzhong, Middle Nianbo and Upper Pana Formations (Mo et al., 2007). The lowest level of the Linzizong Group was dated at ~65 Ma by the $^{40}Ar/^{39}Ar$ method (Mo et al., 2007). Maluski et al. (1982) reported a plateau age ranging from 64.5 to 60.6 Ma for the Dianzhong Formation, based on $^{40}Ar/^{39}Ar$ dating of andesite and plagioclase. Zhou et al. (2004) also reported a plateau age of 64.4 ± 0.6 Ma based on the $^{40}Ar/^{39}Ar$ method, which is similar to the results of Chen et al. (2014). These findings indicate that the age of the Dianzhong Formation is no older than 65 Ma, and that therefore the age of the Shexing Formation lies within the range of ~65 to ~110 Ma.

The sampled locality (29.9°N/91.1°E) lies ~65 km to the northeast of Lhasa city (Fig. 2a). A simplified structural profile of the studied sections is shown in Fig. 2b. The Upper Cretaceous Formation is composed of red sandstones and interbedded basalt flows (at least 11 discrete flows). The thickness of the basalt flows ranges from ~30 cm to 150 cm. The basalt flows directly overlie the red beds with thicknesses ranging from several meters to several tens of meters. In total, we sampled eleven basalt flow sites, and ten Late Cretaceous red bed sites (Figs. 2b, and 3a,b). Each basalt site represents at least one discrete flow. In general, 10 samples were taken at each site, using a gasoline-powered drill. In addition, two fresh basaltic samples were collected at the bottom (LZ10) and the top (LZ19) of the studied section in order to date the whole of the volcanic sequences using the conventional K–Ar method.

3. Methods

3.1. K–Ar dating

K–Ar dating was conducted in the Key Laboratory of Orogenic Belts and Crustal Evolution, Peking University, China. Since the K–Ar method is unable to determine the existence of any extraneous argon, or whether any argon loss has occurred during weathering, we endeavored to collect fresh basalt samples in order to minimize the possibility of these potential sources of error. The processing and measurement procedures used are described in Liu et al. (2004). Fresh rock samples were crushed and sieved using 60–80 mesh sizes. Olivine crystals, which may contain excess argon, were removed using conventional methods in order to minimize any contribution of argon. In addition, dilute nitric acid and conventional methods were used to remove any

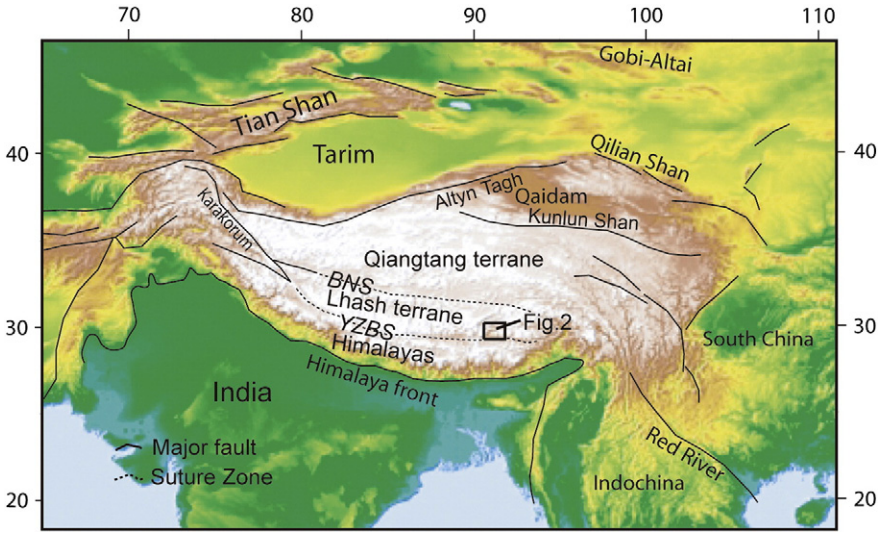


Fig. 1. Simplified tectonic map of Central Asia centered on Tibet, showing blocks and major faults. YZBS: Yaluzangbu Suture, BNS: Bangonghu–Nujiang Suture.

carbonatization and olivine, respectively, in order to minimize their effect on the K–Ar age determinations. Measurements of K content were made using a flame photometer, and argon gas was analyzed using the isotope dilution technique with a mass spectrometer (RGA10).

3.2. Paleomagnetic measurements

Remanent magnetizations were measured using a 2G-755R cryogenic magnetometer at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction of Ministry of Land and Resources, CAGS in Beijing. All samples underwent stepwise thermal demagnetization up to 580 °C or 680 °C and were performed in an ASC TD-48 thermal demagnetizer with internal residual field of less than 10 nT. Remanent magnetizations and thermal demagnetization were carried out in a shielded room with residual fields of less than 300 nT. Demagnetization temperature steps were generally large (40–60 °C) in the low temperature part, and smaller (20–30 °C) at higher temperatures. Magnetization directions were determined by principal component analysis (Kirschvink, 1980), and the site-mean of the paleomagnetic directions was calculated using Fisher statistics (Fisher, 1953).

The anisotropy of magnetic susceptibility (AMS) of 24 red bed samples and 26 basalt samples was measured in order to determine if the rocks have suffered substantive tectonic strain. The measurements were made using a KLY-3 bridge (AGICO).

The magnetic mineralogy of representative lithologies was analyzed using stepwise thermal demagnetization (Lowrie, 1990) of a three-component isothermal remanent magnetization (IRM) using fields of 0.12 T, 0.4 T and 1.2 T for the soft, intermediate and hard components, respectively.

4. Results

4.1. K–Ar dates

The K–Ar analytical data and resulting age determinations are presented in Table 1. The results indicate that the ages of the upper basalt (LZ19) and the lower basalt (LZ10) are 69.75 ± 1.53 Ma and 73.75 ± 1.26 Ma, respectively. Thus, the entire sequence is dated from ~75 Ma to ~68 Ma.

4.2. Paleomagnetic results

The results of thermal demagnetization of a three-component IRM for the red bed samples indicates that the hard, medium and soft components are unblocked at about 680 °C, indicative of hematite (Fig. 4a,b). Thermal demagnetization of the soft fraction of the basalt rocks reveals an unblocking temperature of 580 °C, indicative of magnetite; and the hard and medium fractions exhibit a distinct unblocking temperature of 680 °C, which is probably indicative of hematite in basaltic samples (Fig. 4c,d).

For the red beds, the AMS results reveal that the maximum susceptibility axes (K1) are distributed in a preferred NW–SE orientation within the depositional plane, and the minimum susceptibility axes (K3) are perpendicular to the bedding plane. This results in oblate magnetic ellipsoids (Fig. 5a). The preferred orientation of the K1 axes may be related to NW–SE paleocurrents. The corrected anisotropy degree (p′) of the red beds ranges from 1.004 to 1.03, and their mean anisotropy degree is 1.016. The basalts exhibit a larger mean anisotropy degree of 1.019. For the red beds and basalts, the anisotropy shape factor (T) varies widely, independently of p′, and is randomly distributed between prolate and oblate forms. For the basalts, the principal anisotropy directions are dispersed (Fig. 5b). Based on the low degrees of anisotropy, we suggest that it is unlikely that either the basalts or the red beds have experienced significant strain due to compaction or tectonic stresses. In addition, representative samples were surveyed using a light microscope. Reflected light microscopy conducted on representative samples showed that the mineral grains are interspersed in a matrix groundmass of basalt, and some crystals are tear-shaped (Fig. 3c, d), suggesting the typical rapidly cooled texture of volcanic rocks. In addition, microscopic observation through polished thin sections indicated an essentially original magmatic texture, with no chemical alteration caused by fluids or weathering in the groundmass (Fig. 3c,d). Therefore, the magnetic grains of our samples are likely to originate from the time of rock formation so that the Characteristic Remanent Magnetization (ChRM) carried by them is probably the primary magnetization.

Representative Zijderveld plots are illustrated in Fig. 6. The demagnetization behavior of the red beds is similar to that of the basalt samples. Two components could be separated in most of the basalt and red bed samples: a lower-temperature component (LTC), which was removed below 300 °C, and a higher-temperature component (HTC), isolated between 300 °C and 580 °C or 680 °C (Fig. 6). In some cases, only a HTC could be separated. The high unblocking temperature (580 °C or 680 °C) also indicates the presence of magnetite or hematite for the red bed and basalt sites.

The site-mean directions are listed in Table 2 and illustrated in Fig. 7. The Characteristic Remanent Magnetization (ChRM) directions in geographic coordinates are significantly different from the present-day field direction. The site mean HTC direction for the volcanic rocks is

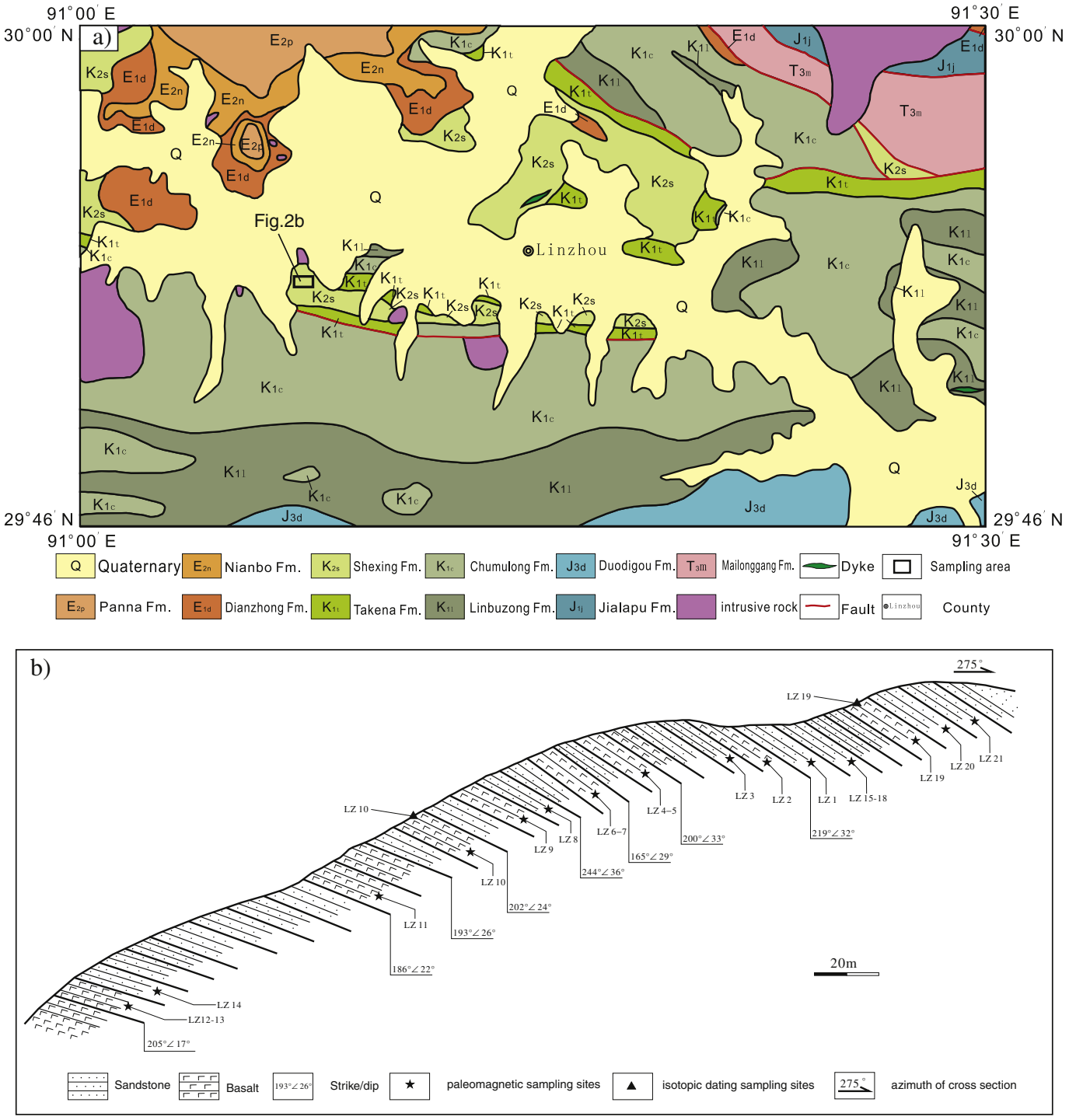


Fig. 2. (a) Simplified geological map of the investigation area; (b) and measured geological section of the Shexing formation near Linzhou county, showing details of paleomagnetic sampling locations, site distributions, and strike direction and dip azimuths for each sampling site.

Dg = 11.1°, Ig = 30.6°, kg = 32.4 (α_{95} = 8.1°, n = 11, in geographic coordinates) (Ds = 359.4°, Is = 22.5°, ks = 18.9, α_{95} = 10.8°, n = 11, after tectonic correction), and the corresponding paleopole lies at 71.8°N, 273.0°E with dp/dm = 6.1/11.4°. The site mean ChRM direction for the red beds is Dg = 15.6°, Ig = 32.6°, kg = 39.7 (α_{95} = 7.8°, n = 10, in geographic coordinates) (Ds = 1.7°, Is = 17.9°, ks = 38.1, α_{95} = 7.9°, n = 10, after tectonic correction), and the corresponding paleopole lies at 69.2°N, 266.4°E with dp/dm = 4.3/8.2°. The consistent inclination values recorded in the red beds and lava flows implies no significant inclination shallowing caused by deposition compaction in red beds from the studied section.

For the volcanic rocks, the reversal test (McFadden and McElhinny, 1990) angle between the mean directions produced a positive result ($r = 19.1° < r_{critic} = 19.5°$, classification C). For the red beds, the McFadden (1990) correlation fold tests yielded Xi2 IS = 4.265, Xi2 TC = 0.827, critic Xi at 95% = 3.685, in geographic and stratigraphic coordinates, respectively; and therefore the result is also positive. Thus, we suggest that the remanence is probably primary. The mean

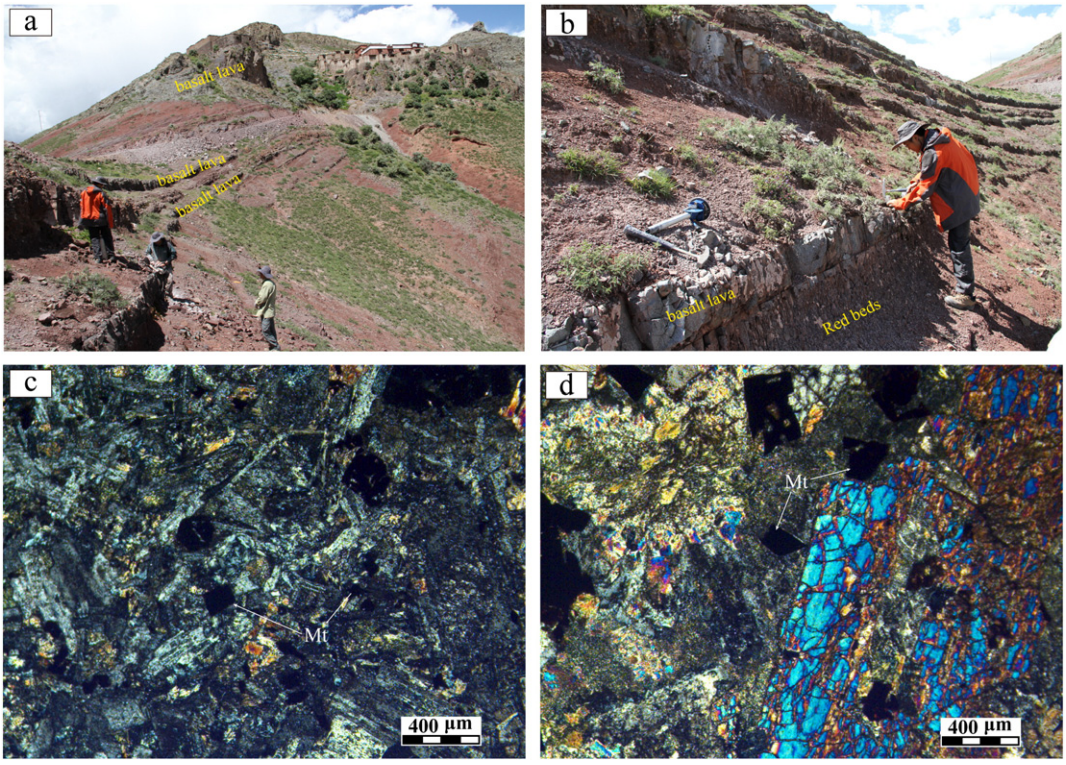


Fig. 3. (a–b) show the field outcrops of the Upper Cretaceous Shexing Fm red beds and interbedded volcanic rocks; (c–d) Microscopic observation results for samples from the volcanic rocks. Mt: magnetite.

inclination of the red bed sites (17.9° ± 7.9°) is similar to that of the basalt series (22.5° ± 10.8°) within the limits of statistical uncertainty (Fig. 7). This implies that the inclinations recorded in red beds are not affected by inclination shallowing due to compaction with respect to that of basalts during the Late Cretaceous. Hence, the mean directions of the basalt series and red beds sites can be averaged together (Ds = 0.5°, Is = 20.2°, κs = 25.5, α_{95} = 6.4°, N = 21), with a corresponding paleopole at 70.5°N, 269.6°E with A_{95} = 4.9°, dp/dm = 3.5/6.7°. Since our site-mean directions are based on 11 independent basalt flows, each basalt lava site represents at least one discrete flow. Consequently, the mean of the 21 sites from the studied section almost certainly provides adequate averaging of the geomagnetic field. We tested the angular dispersion of the virtual geomagnetic poles (VGP) and obtained a VGP scatter of 13.2°, which falls within the limit (10.4°–13.9°) of the predicted VGP (McFadden et al., 1991) for 0–20° latitude for the interval from 45 to 80 Ma. This suggests that the overall mean direction of the HTC from the basalt flows may have fully averaged out the secular variation at the time during emplacement of the flows and is an accurate estimate of the time-averaged (68–75 Ma) paleomagnetic pole.

Based on the analysis above, we suggest that the paleomagnetic direction from the Late Cretaceous volcanics and red beds can be regarded as the primary magnetization. Thus, these ChRM directions are suitable for tectonic reconstructions.

5. Discussion

5.1. The pre-collisional paleolatitudinal positions of the southern margin of Asia

Published paleomagnetic data from sedimentary rocks and lavas of Cretaceous to Paleogene age from the Lhasa terrane have been used to construct a detailed paleolatitude record for southern Tibet. However, the results from these studies are highly variable, with calculated paleolatitudes ranging from 8°N to 33°N (Achache et al., 1984; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Tan et al., 2010). In addition, several authors assume a collision age of less than 45–34 Ma at higher paleolatitudes, greater than 22°N (Aitchison et al., 2007; Tan et al., 2010; Dupont-Nivet et al., 2010a). Based on either igneous (Liebke et al., 2010; Sun et al., 2010) or sedimentary (Sun et al., 2012) rock formations from the Lhasa terrane, these paleomagnetic studies imply lower paleolatitudes (10°–15°N) and older ages (49–65 Ma) for the collision, as proposed earlier by Besse et al. (1984) and Patzelt et al. (1996). Using a new APWP for East Asia, Cogné et al. (2013) proposed that the Lhasa paleolatitude was situated at 13.2 ± 8.5°N south of East Asia when the collision between India and Asia occurred prior to 50 Ma.

Published Cretaceous to Paleogene studies from the Lhasa block are listed in Table 3, and the corresponding paleopoles are shown in Fig. 8.

Table 1
Detailed results of K–Ar dating of basaltic rocks from the studied section.

Sample ID	Age (Ma)	±	K %	Weight (g)	Exp Serial	Radiogenic ^{40}Ar		^{38}Ar (10^{-11} mol)	^{40}Ar/^{38}Ar	^{40}Ar/^{38}Ar (±)	^{36}Ar/^{38}Ar	^{36}Ar/^{38}Ar (±)	^{40}Ar*/^{40}K	^{40}Ar*/^{40}K (±)
						mol/g	%							
LZ10-6	73.75	1.26	0.5	0.0248	4220	6.53E−11	42.33	1.96E−11	0.1955081	0.000181	0.0003920	0.0000004	0.004374	0.000076
LZ19-2	69.75	1.53	0.29	0.0254	4223	3.58E−11	25.75	1.96E−11	0.1804006	0.0003261	0.0004649	0.0000002	0.004132	0.000092

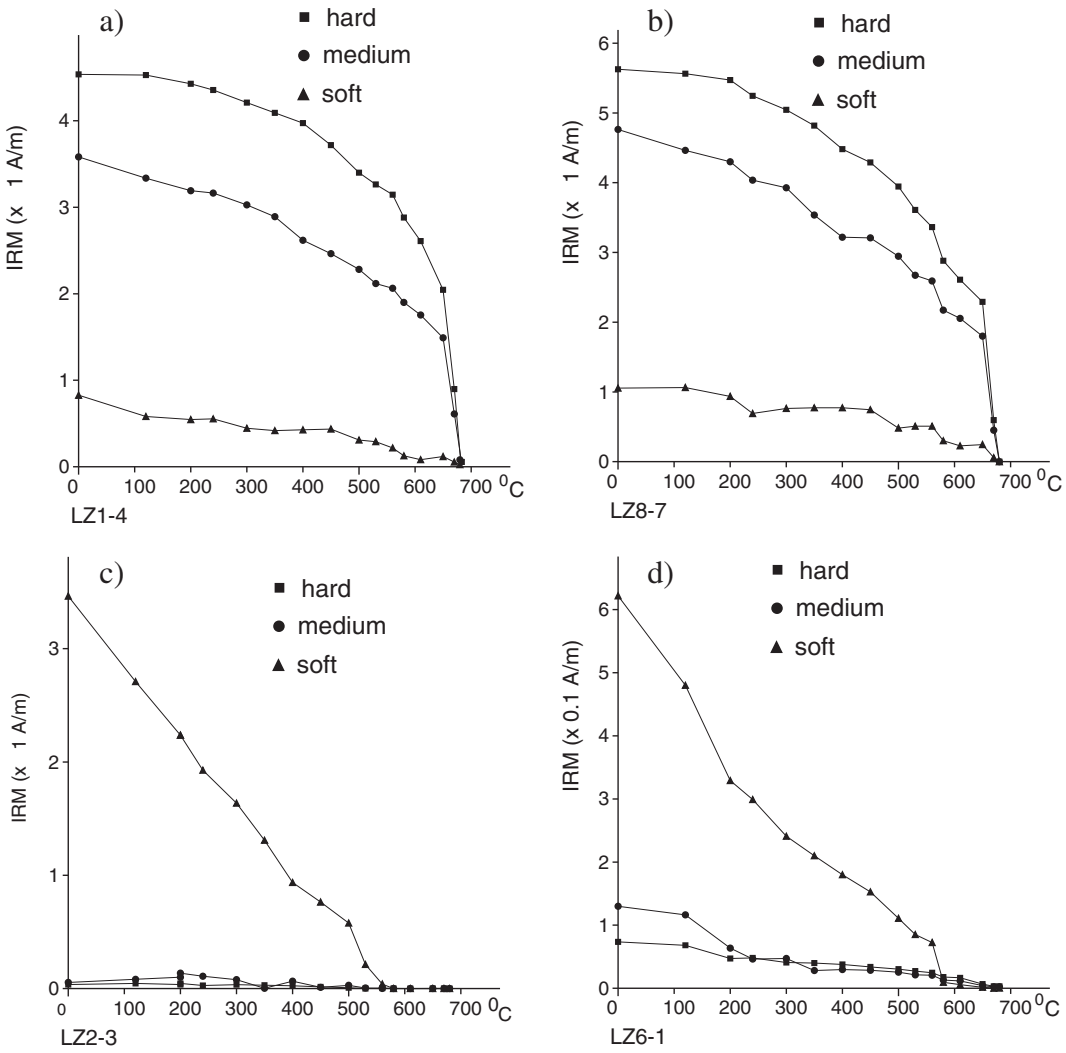


Fig. 4. Thermal demagnetization of a three-component IRM of representative samples, showing unblocking temperatures around 580 °C or 680 °C. The soft (<0.12 T), medium (0.12–0.4), and hard (0.4–1.2 T) components are shown during thermal demagnetization. (a–b) Typical examples from red beds; (c–d) Typical examples from volcanic rocks.

The corresponding paleolatitudes have been calculated for a reference point in our studied section (29.9°N, 91.1°E, Linzhou County). Most of the Cretaceous paleomagnetic data reveal that the Lhasa terrane was located at a low paleolatitudinal position (~10°N to ~15°N) during the Early and Late Cretaceous (Achache et al., 1984; Lin and Watts, 1988; Chen et al., 1993; Sun et al., 2008, 2012; Chen et al., 2012; Tang et al., 2013; Li et al., 2017; Yang et al., 2015; Yi et al., 2015). Ma et al. (2014) estimated a paleolatitude of 20.7° ± 2.1°N of the Lhasa terrane during the Early Cretaceous; and Tan et al. (2010) reported a higher paleolatitude of 24.1° ± 4.2°N of the Lhasa terrane during the Late Cretaceous. For the Paleogene, several researchers have also suggested a low paleolatitudinal position (~10°N to ~15°N) of the Lhasa terrane (Achache et al., 1984; Chen et al., 2010; Liebke et al., 2010; Sun et al., 2010; Chen et al., 2014). However, other studies have yielded a higher paleolatitude of ~22 to ~32°N (Tan et al., 2010; Dupont-Nivet et al., 2010a; Meng et al., 2012; Huang et al., 2013, 2015a).

As revealed in Fig. 8, some of the paleopoles for the Lhasa terrane differ significantly from those of other studies, and these differences among Late Cretaceous poles discourage us from using the average of the currently available poles as a reference for the entire Lhasa terrane. Consequently, we applied the sifting approach towards paleomagnetic data reported by Van der Voo (1990) in order to assign a Q-value quality factor to each result (Table 3). Most of the Late Cretaceous paleomagnetic data for the Lhasa terrane have a Q-value of five or greater (Table 3). The pole calculated from Lin and Watts (1988) had a low Q-value. Although Tan et al. (2010) reported paleomagnetic data from Late Cretaceous red beds interbedded with lava flows, a very large difference exists in the declination and inclination values recorded in the red beds and lava flows. Hence, the three poles are excluded from the subsequent discussion. The other Late Cretaceous poles of the Lhasa terrane have been used to constrain the paleolatitude of the Lhasa terrane during the Late Cretaceous. The Late Cretaceous poles from similar regions in the Lhasa terrane are aligned on a well-defined small circle centered on the reference point (29.9°N, 91.1°E). This indicates that the paleolatitude for the Lhasa terrane was constant during the Late Cretaceous. The small circle that passes over the ten poles (Pozzi et al., 1982; Westphal and Pozzi, 1983; Achache et al., 1984; Lin and Watts, 1988; Sun et al., 2012; Tang et al., 2013; Yang et al., 2015; Yi et al., 2015; this study) is fitted by a reference point near the Linzhou area (29.9°N, 91.1°E) with a colatitude of 75.9° ± 3.3°. Subsequently, the Late Cretaceous paleolatitude of 14.1° ± 3.3° N for the Lhasa terrane was calculated at the same reference point (Fig. 8a).

The same analysis was also applied to the Paleogene paleopoles of the Lhasa terrane (Fig. 8b). All of the Paleogene paleomagnetic data for the Lhasa terrane have a Q-value of five or greater (Table 3). The paleopole reported by Huang et al. (2013) had no robust field test to assure data reliability. The Eocene pole proposed by Tan et al. (2010) differs significantly from that of other studies of the Lhasa terrane.

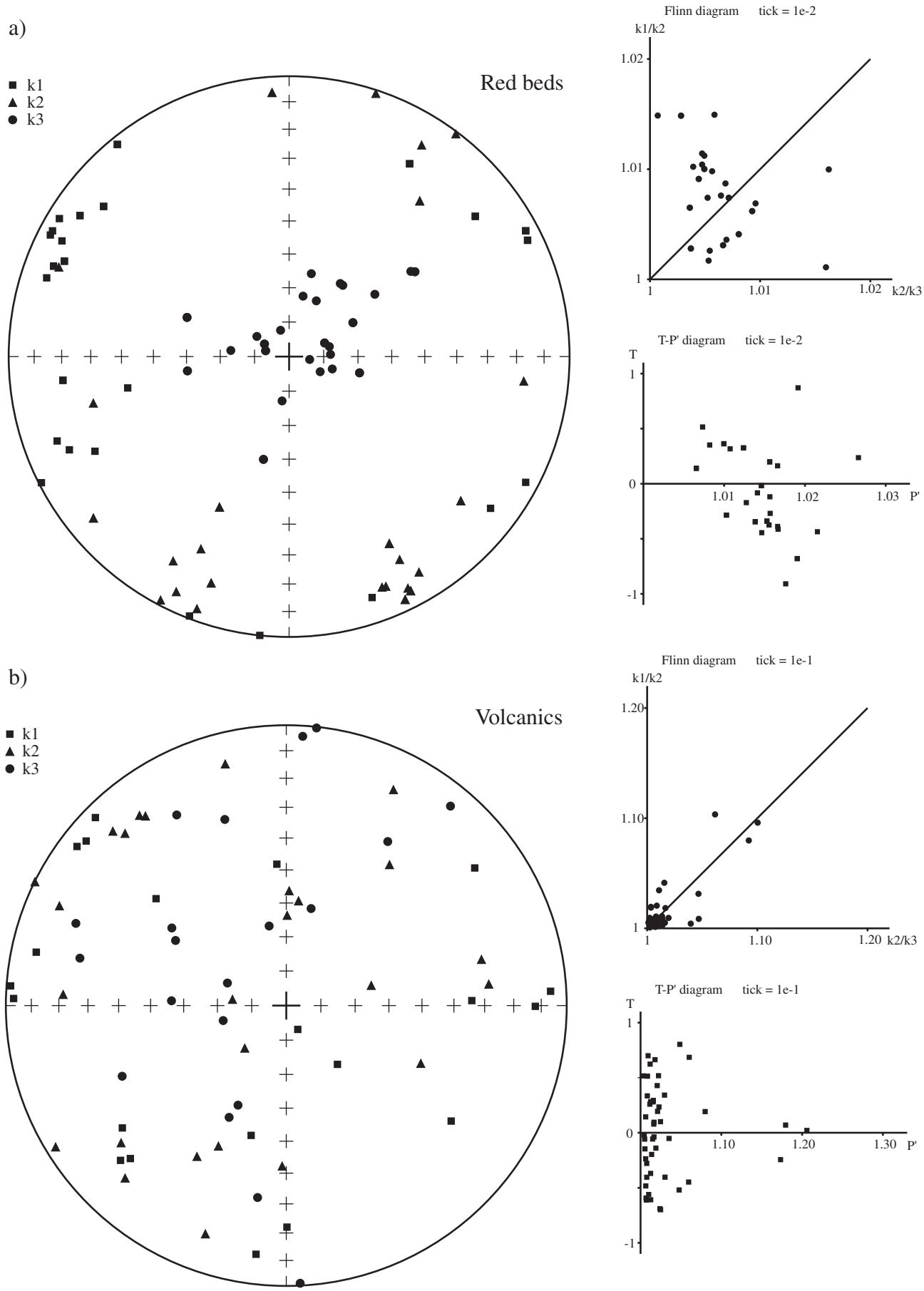


Fig. 5. Anisotropy of magnetic susceptibility data in stratigraphic coordinates from the Shexing formation rocks. Lineation = $k1/k2$, Foliation = $k2/k3$. (a) Red beds; (b) volcanic rocks.

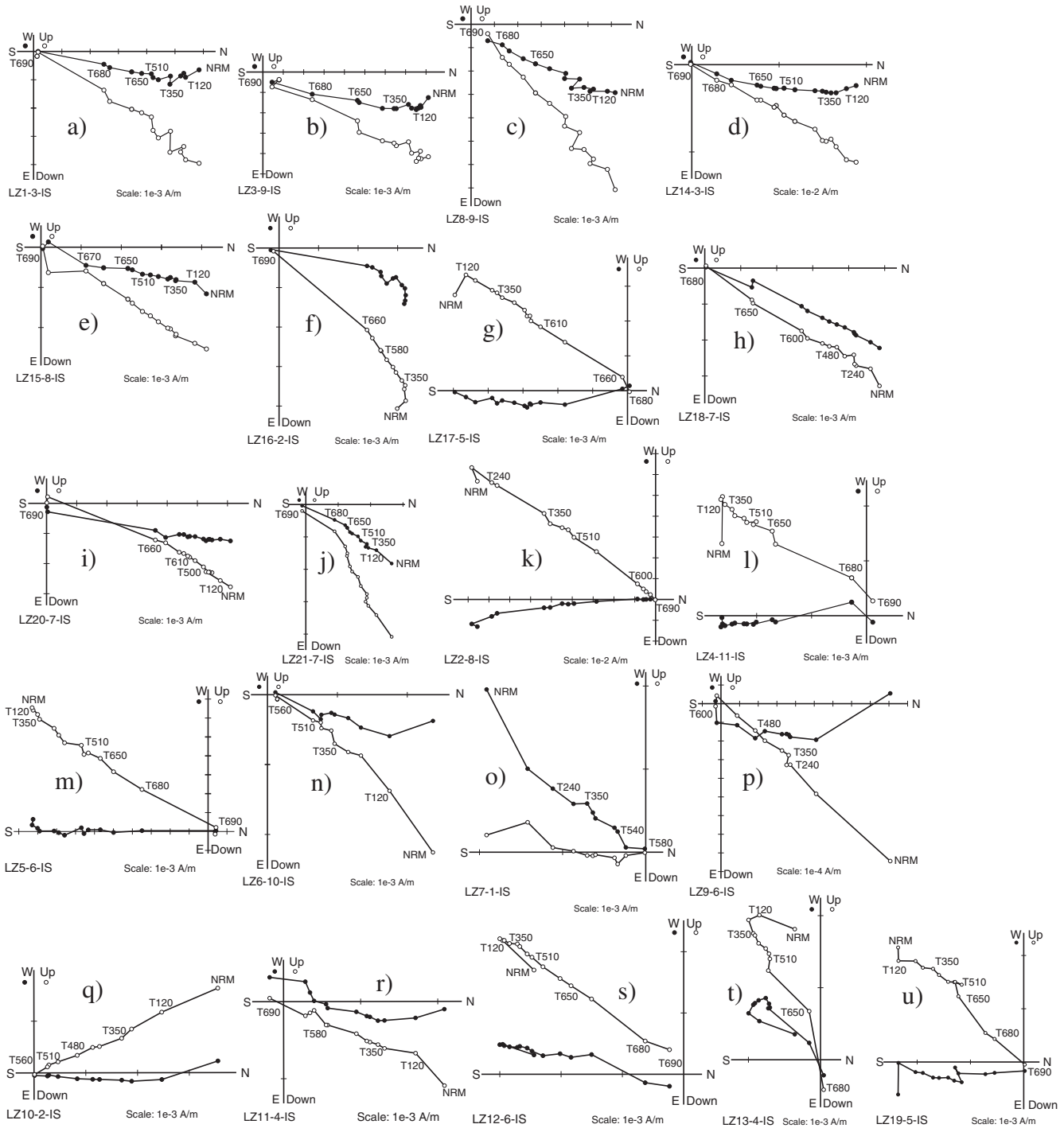


Fig. 6. Orthogonal demagnetization diagrams of representative rocks (sample number and corresponding sites listed) in situ coordinates. Demagnetization steps in °C in all plots. NRM is the natural remanent magnetization. (a–j) Typical examples from red beds; (k–t) Typical examples from volcanic rocks.

Apart from the Paleocene poles from Huang et al. (2013) and Tan et al. (2010), the other Paleocene poles from similar regions in the Lhasa terrane are aligned on a well-defined small circle centered on the reference point (29.9°N, 91.1°E). This also suggests a constant paleolatitude for the Lhasa terrane during the Paleocene. The small circle that passes through the eight poles (Achache et al., 1984; Chen et al., 2010; Dupont-Nivet et al., 2010a; Liebke et al., 2010; Sun et al., 2010; Meng et al., 2012; Chen et al., 2014) is fitted by a reference point near the Linzhou area (29.9°N, 91.1°E) with a colatitude of 75.6° ± 4.8°. Subsequently, the Paleogene paleolatitude of 14.4° ± 4.8°N for the Lhasa terrane was calculated at the same reference point (Fig. 8b).

Based on the results of K–Ar dating and paleomagnetic analysis presented here, we obtained a new Late Cretaceous paleopole (68–75 Ma) for the Lhasa terrane of 70.5°N, 269.6°E with $A_{95} = 4.9°$, which indicates that the Lhasa terrane was located at a paleolatitude of 10.4°N ± 4.9° around 70 Ma (reference point: 29.9°N, 91.1°E). Combining our results with previous paleomagnetic data from the Lhasa terrane, we conclude that the Lhasa terrane experienced no paleomagnetically-detectable northward movement prior to the Cenozoic India–Asia collision and remained essentially stationary at ~10 to ~15°N during the Cretaceous to the Paleocene. This paleolatitude is in agreement with some other Cretaceous and Paleogene paleomagnetic results from the Lhasa terrane

Table 2
Site-mean paleomagnetic directions of the high-temperature from Late Cretaceous rocks in Linzhou Basin (29.9°N, 91.1°E), Northern Lhasa.

Site	Slat (N)	Slon (E)	Strike/dip (°)	−/+	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95} (°)	Plat (°)	Plon (°)
Red beds												
LZ1	29°56.889′	91°11.922′	219/32	0/11	14.9	34.3	1.2	17.3	76.1	5.3	68.9	267.8
LZ3	29°56.901′	91°11.933′	200/33	0/6	19.0	31.4	0.9	25.5	92.3	7.0	73.5	268.0
LZ8	29°56.887′	91°11.933′	211/33	0/8	8.3	39.7	350.9	22.0	39.9	8.9	69.7	297.6
LZ14	29°56.857′	91°11.020′	205/20	0/10	12.6	20.2	6.4	14.9	98.6	4.9	66.9	254.8
LZ15	29°56.931′	91°11.931′	208/19	0/7	13.3	25.8	5.7	19.7	65.6	7.5	69.5	254.9
LZ16	29°56.911′	91°11.904′	223/36	1/2	16.6	45.0	356.4	22.8	63.4	15.6	71.7	282.4
LZ17	29°56.911′	91°11.904′	223/36	5/0	169.7	−31.5	162.2	−3.1	23.2	16.7	57.0	305.2
LZ18	29°56.938′	91°11.913′	220/45	0/11	15.3	29.2	7.8	2.5	24.0	9.6	60.4	255.2
LZ20	29°56.938′	91°11.913′	215/36	0/8	24.7	28.9	9.5	17.4	19.6	13.0	67.2	246.2
LZ21	29°56.945′	91°11.908′	215/36	0/10	41.6	34.4	17.7	30.8	6.3	20.9	69.0	216.6
Mean	29°56.945′	91°11.908′		10	15.6	32.6	–	–	39.7	7.8		
					–	–	1.7	17.9	38.1	7.9		
Corresponding pole: 69.2°N/266.4°E, d_p/d_m = 4.3°/8.2°												
Basalt												
LZ2	29°56.880′	91°11.921′	167/32	4/3	177.5	−14.5	168.0	−17.8	14.0	11.6	66.4	301.9
LZ4	29°56.897′	91°11.933′	165/25	4/5	182.5	−33.7	161.4	−37.4	93.2	5.4	71.0	337.2
LZ5	29°56.897′	91°11.933′	165/25	5/2	183.7	−25.3	168.1	−30.9	20.7	9.4	72.9	313.2
LZ6	29°56.887′	91°11.933′	244/36	3/4	16.3	27.5	10.7	−0.6	17.2	9.9	58.1	250.5
LZ7	29°56.887′	91°11.933′	244/36	1/4	21.2	29.6	13.7	3.0	41.6	12.0	58.8	243.9
LZ9	29°56.890′	91°11.938′	202/24	0/9	23.7	37.0	6.3	34.0	36.6	6.0	77.4	242.7
LZ10	29°56.869′	91°11.938′	193/26	9/0	185.8	−13.9	180.4	−9.4	13.5	10.1	64.8	270.2
LZ11	29°56.862′	91°11.940′	186/22	0/7	15.5	18.9	7.5	21.0	46.7	6.2	69.7	249.4
LZ12	29°56.867′	91°11.020′	205/17	9/0	207.1	−43.5	191.3	−41.8	14.1	14.2	78.4	208.4
LZ13	29°56.867′	91°11.020′	205/17	5/1	191.5	−46.7	178.5	−39.3	10.0	14.5	82.2	281.4
LZ19	29°56.938′	91°11.913′	220/45	4/6	1.7	42.2	346.0	8.7	24.0	6.7	61.2	301.2
Mean	29°56.938′	91°11.913′		11	11.1	30.6	–	–	32.4	8.1		
					–	–	359.4	22.5	18.9	10.8		
Corresponding pole: 71.8°N/273.0°E, d_p/d_m = 6.1°/11.4°												
Mean	Red beds + basalt			21	13.2	31.6	–	–	36.4	5.3		
					–	–	0.5	20.2	25.5	6.4		
Corresponding pole: 70.5°N/269.6°E, A_{95} = 4.9°												

Slat and slon: latitude and longitude of the sampling site; Strike: dip direction of beds - 90°, Dip: the mean dip and dip azimuth of site beds; −/+, number of ChRM directions that display reversed/normal polarity in calculating Fisher site-means; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigraphic coordinates respectively; κ: the best estimate of the precision parameter; α_{95}: the radius that the mean direction lies within 95% confidence; Plat and Plon, latitude and longitude of pole.
(1) Fold test for red beds is positive according to McFadden method (1990): critic Xi at 95% = 3.685, Xi2 IS = 4.625, Xi2 TC = 0.827.
(2) Reversal test for basalt is positive: angle between the mean directions r = 19.1° < r_{critic} = 19.5°, classification C (McFadden and McElhinny, 1990).

(Achache et al., 1984; Chen et al., 1993; Sun et al., 2008, 2010; Chen et al., 2010; Liebke et al., 2010; Chen et al., 2012; Sun et al., 2012; Tang et al., 2013; Chen et al., 2014; Ma et al., 2014; Li et al., 2017; Yang et al., 2015; Yi et al., 2015).

With regard to the Himalayan terrane, Patzelt et al. (1996) obtained reliable paleomagnetic results from the Zongshan Formation and the Zongpu Formation which yielded paleolatitudes of 5.4 ± 3.8°N (63–55 Ma) and 3.9 ± 4.4°S (71–65 Ma) for the reference point of 29.9°N, 91.1°E (Table 3). Yi et al. (2011) carried out a paleomagnetic study of Paleocene marine sediments in the Gamba area of the Tethyan Himalayan terrane, which yielded two high quality paleopoles with paleolatitudes of 7.3 ± 3.5°N (62–59 Ma) and 11.6 ± 2.5°N (59–56 Ma) (reference point: 29.9°N, 91.1°E) (Table 3), indicating that the Himalayan terrane moved northward at a velocity of 15.1 cm/yr prior to the collision. Extrapolating the constant northward drift velocity of 15.1 cm/yr for the Himalayan terrane, it would take 2.0 ± 3.8 Ma for its northern margin to overlap with the southern margin of the Lhasa terrane at 14.4 ± 4.8°N. Therefore, we favor an initial age for the collision of the Himalayan terrane and Lhasa terrane of 54 ± 3.8 Ma, which generally accords with previous paleomagnetic results (Sun et al., 2012; Ma et al., 2014; Yang et al., 2015) and geological studies (Gaetani and Garzanti, 1991; Guillot et al., 2003, 2007, 2008; Copley et al., 2010; Najman et al., 2010).

5.2. Insights into post-collisional convergences within Asia

Tectonic activities within the study region (e.g. intracontinental shortening between the southern margin of the Lhasa terrane and the Qiangtang terrane and in the Kunlun Shan, Qilian Shan, Tien Shan, Altai, and other mountain ranges; intracontinental subduction beneath Qaidam and Tibet; and strike-slip faulting along the Altyn Tagh, Karakorum) are the consequence of the continental collision between India and Asia. Yin and Harrison (2000) reported that at least 1400 km of north–south shortening has been absorbed by the orogenic belts since the onset of the India–Asian collision. Based on a mass-balanced cross-section, DeCelles et al. (2002) and Guillot et al. (2003) produced estimates of total Himalayan shortening of 670 km and 1100 km, respectively. Johnson (2002) proposed a post-collisional shortening of 1400 km between the Himalayan and Tibetan orogen. The Lhasa, Qiangtang and Songpan–Ganze terranes of the Tibetan Plateau were significantly shortened through folding and thrusting during the Cretaceous to Paleogene. The Qiangtang terrane underwent ~300 km shortening during the interval of 140–100 Ma during the Lhasa–Qiangtang collision (DeCelles et al., 2007). Subsequently, geological studies have documented 600–750 km of N–S Cenozoic shortening across the northern part of the Tibetan Plateau. At the longitude of central Tibet, ~450 km of Tibetan shortening, an ~150 km N–S shortening component associated with the ATF, and ~150 km of shortening in the Tien Shan and Mongolia, together result in ~750 km of shortening since 50 Ma. In western Tibet, south of the Pamir-Hindu Kush region, the total amount of intra-Asian shortening is ~1050 km since 50 Ma. The Lhasa terrane underwent ~250 km shortening from 100 to 50 Ma (Murphy et al., 1997; Kapp et al., 2005; Guynn et al., 2006; Kapp et al., 2007; Volkmer et al., 2007). Estimates of Cenozoic shortening by thrusting in the southern Qiangtang terrane and along the Bangong–Nujiang suture zone (from 30 to 20 Ma) are ~120 km in total (Kapp et al., 2005, 2007; Lippert et al., 2011).

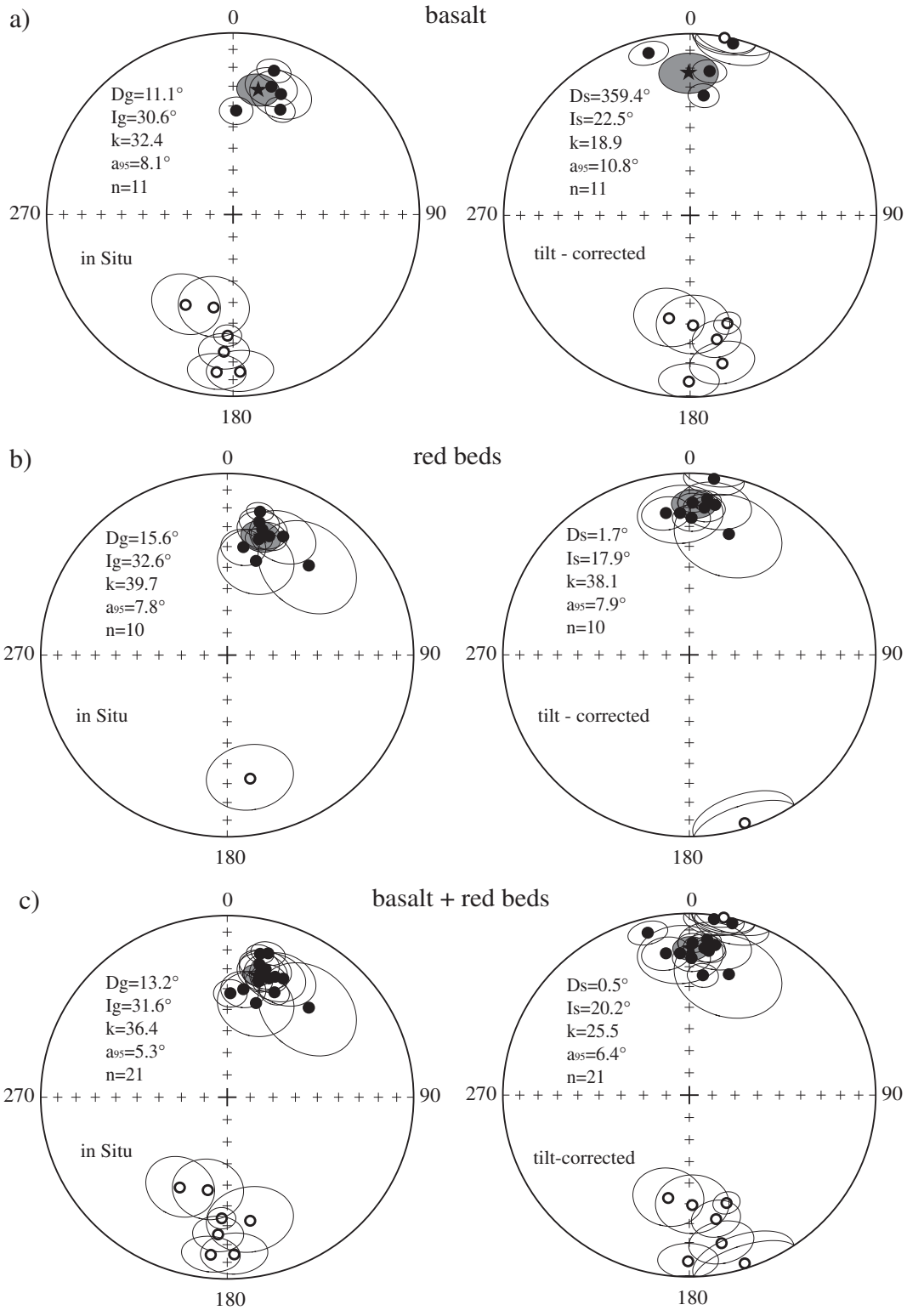


Fig. 7. Equal-area projections of the high temperature component for volcanic rocks (a) and for red beds (b), and all sites including volcanic rocks (c).

Numerous paleomagnetic studies and syntheses have proposed various estimates of the magnitude of both the total pre-collisional Lhasa/Siberia shortening and its distribution throughout Tibet, in Lhasa, Qiangtang and in the Qaidam blocks and the reactivated mountain chains such as the Kunlun, Tien Shan and Altai ranges. Earlier paleomagnetic studies and syntheses (Besse et al., 1984; Patriat and Achache, 1984; Patzelt et al., 1996) have suggested a total convergence of India and Siberia on the order of 2700–3000 km, and up to 3500 km, using the European reference APWP (Besse and Courtillot, 2002). Using a new reference APWP for eastern Asia, Cogné et al. (2013) propose that the total convergence is 2080 ± 630 km between India and Siberia through Greater India and Central Asia since 50 Ma, and that a post-Cretaceous convergence amounts to 1250 ± 1020 km between the southern parts of Tibet and northernmost Siberia. These values

Table 3
Paleomagnetic poles from Lhasa terrane and from Himalayan terrane.

Age (Ma)	Rock type	Slat. (°N)	Slon. (°E)	N	Plat. (°N)	Plon. (°E)	A_{95} (dp/dm)	Paleolat. (°N)	Criteria (Q)	Reference
The Lhasa terrane										
132–120	Volcanic rocks	32.3	82.6	51	61.4	192.9	2.1	20.7 ± 2.1	123R5D7(7)	Ma et al. (2014)
130–110	Volcanic rocks	31.3	85.1	18	58.2	341.9	4.6	15.9 ± 4.6	123F5R7(7)	Chen et al. (2012)
~120	Volcanic rocks + sediments	31.3	91.9	19	66.9	281.2	6.1	7.1 ± 6.1	123F5D□(7)	Li et al. (2017)
117–121	Volcanic rocks	31.1	81.4	12	70.5	292.9	7.4	11.6 ± 7.4	123□5D7(6)	Yang et al. (2015)
K_1	Volcanic rocks	30.5	90.1	15	66.4	220.3	6.9	13.7 ± 6.9	123□5□7(5)	Sun et al. (2008)
K	Sediments	33.7	80.4	14	66.2	245.0	5.1	8.2 ± 5.1	123F5D7(7)	Chen et al. (1993)
99–93	Volcanic rocks	30.9	85.1	14	63.1	224.6	5.1	10.1 ± 5.1	123F5□7(6)	Tang et al. (2013)
~96	Volcanic rocks	31.5	92.0	9	78.0	282.0	4.0/6.9	18.1 ± 4.0	123□□□(3)	Lin and Watts (1988)
~92	Volcanic rocks	32.4	80.1	10	64.1	209.0	9.6	15.7 ± 9.6	123F5R7(7)	Yi et al. (2015)
~90	Volcanic rocks	31.7	90.9	4	74.0	318.0	11.1/19.1	18.4 ± 11.1	1□3□5□□(3)	Lin and Watts (1988)
~80	Volcanic rocks	31.6	82.2	15	68.4	298.8	2.7	10.4 ± 2.7	123F5□7(6)	Yi et al. (2015)
K_2	Red beds	31.2	84.7	33	49.0	344.3	5.3	12.2 ± 5.3	123F5D7(7)	Yang et al. (2015)
K_2	Sediments	29.9	91.2	43	70.2	300.5	1.9	12.3 ± 1.9	123F5D7(7)	Tan et al. (2010)
K_2	Volcanic rocks	29.9	91.2	21	69.1	191.7	4.2	24.1 ± 4.2	123F5□7(6)	Tan et al. (2010)
K_2	Sediments	29.9	91.2	8	68.0	279.0	3.5/6.9	8.1 ± 3.5	123F5□□(5)	Lin and Watts (1988)
K_2	Sediments	30.0	91.0	8	71.2	288.4	7.9	11.8 ± 7.9	123F5□□(5)	Achache et al. (1984)
K_2	Sediments	29.9	91.0	6	64.0	348.0	5.6/9.5	21.2 ± 5.6	123F5□7(6)	Westphal and Pozzi (1983)
K_2	Sediments	29.4	91.0	7	68.0	340.0	6.0	20.2 ± 6.0	123F5□7(6)	Pozzi et al. (1982)
K_2	Volcanic rocks + sediments	29.9	90.7	20	75.0	306.7	6.8	17.4 ± 6.8	123F5D7(7)	Sun et al. (2012)
~70	Volcanic rocks + sediments	29.9	91.1	21	70.5	269.6	4.9	10.4 ± 4.9	123F5R7(7)	This study
65–45	Volcanic rocks	30.0	91.0	8	71.4	299.8	11.0	13.3 ± 11.0	123F5D□(6)	Achache et al. (1984)
64–60	Volcanic rocks	30.0	91.2	20	66.4	262.5	6.3	6.5 ± 6.3	123F5D7(7)	Chen et al. (2010, 2014)
60–50	Volcanic rocks	30.0	91.2	13	69.7	268.6	6.3	9.6 ± 6.3	123F5D7(7)	Chen et al. (2010, 2014)
50–44	Volcanic rocks	30.0	91.2	18	69.1	234.2	5.6	12.6 ± 5.6	123F5D7(7)	Chen et al. (2010, 2014)
57–54	Sediments	29.9	84.3	–	78.0	329.0	5.9	23.1 ± 5.9	123F5□7(6)	Meng et al. (2012)
~55	Volcanic rocks	30.1	90.9	22	73.2	282.4	5.4	13.4 ± 5.4	123F5D7(7)	Sun et al. (2010)
54–43	Volcanic rocks + sediments	30.0	91.2	–	68.4	243.0	2.0	10.5 ± 2.0	123□5□7(5)	Huang et al. (2013)
54–47	Volcanic rocks	30.0	91.1	24	76.4	212.6	5.2	22.2 ± 5.2	123F5D7(7)	Dupont-Nivet et al. (2010a)
~53	Volcanic rocks	30.1	90.9	10	69.0	225.3	7.8	14.4 ± 7.8	123F5□7(6)	Liebke et al. (2010)
43–40	Volcanic rocks	29.9	91.2	10	87.5	81.4	5.9	32.4 ± 5.9	123F5□7(6)	Tan et al. (2010)
Oligocene	Sediments	32.2	84.3	35	64.8	314.3	2.6	10.5 ± 2.6	123F5R7(7)	Ding et al. (2014)
The Himalayan terrane										
62–59	Sediments	28.3	88.5	18	67.3	266.3	3.5	7.3 ± 3.5	123F5□7(6)	Yi et al. (2011)
59–56	Sediments	28.3	88.5	14	71.6	277.8	2.5	11.6 ± 2.5	123F5□7(6)	Yi et al. (2011)
71–65	Sediments	28.3	88.5	14	55.8	261.4	4.4/8.6	−3.9 ± 4.4	123F5□7(6)	Patzelt et al. (1996)
63–55	Sediments	28.3	88.5	14	65.4	277.6	3.8/7.6	5.4 ± 3.8	123□5□7(5)	Patzelt et al. (1996)
APWP of the India craton										
55					60.7	277.7	4.8	0.8 ± 4.8		Besse and Courtillot (2002)
70					36.8	281.0	4.5	−22.7 ± 4.5		Besse and Courtillot (2002)
APWP of the East Asia										
50					81.6	238.8	5.9	22.7 ± 5.9		Cogné et al. (2013)
70					79.7	219.5	3.3	23.2 ± 3.3		Cogné et al. (2013)

Notes: Slat. and Slon., latitude and longitude of the sampling location; N, number of samples or sites used to calculate Fisher mean; Plat. and Plon., latitude and longitude of the pole; A_{95}, the radius that the mean pole lies within 95% confidence; dp/dm, semi-axes of elliptical error of the pole at a probability of 95%; Paleolat., paleolatitude calculated for the reference point at 29.9°N, 91.1°E; Criteria (Q), data quality criteria (number of criteria met) after Van der Voo (1990) [1, well determined rock age; 2, sufficient number of samples, N ≥ 24, *k* (or *K*) ≥ 10 and α_{95} (A_{95}) ≤ 16.0; 3, systematic step-wise demagnetization; 4, field tests: F means positive fold test; 5, structural control and tectonic coherence with the craton or terrane discussed; 6, presence of reversal: R means positive reversal test and D means dual-polarity ChRM direction; 7, no resemblance to paleopole of younger age (by more than a period); "□" in the criterion column fails to fulfill this criterion].

are slightly lower than the previously published estimates of 1500–2000 km for the post-Cretaceous intra-continental shortening between southern Tibet and Siberia (Achache et al., 1984; Lin and Watts, 1988; Patzelt et al., 1996; Liebke et al., 2010; Sun et al., 2010; Dupont-Nivet et al., 2010a; Sun et al., 2012; Van Hinsbergen et al., 2012; Tang et al., 2013; Li et al., 2017), but are similar to the results of Ma et al. (2014), Yang et al. (2015) and Yi et al. (2015). However, several researchers suggest much smaller amounts of continental shortening (only a few hundred km) within Tibet (Tan et al., 2010; Chen et al., 2012; Meng et al., 2012). The large data scatter is obvious, but it does not clearly appear to pertain to the magnetization acquisition mechanisms, even though some of the results from sedimentary rocks may have been affected by inclination shallowing, as discussed by Tan et al. (2003, 2010).

Previous studies (Cogné et al., 1999; Tan et al., 2003, 2010; Dupont-Nivet et al., 2010b; Najman et al., 2010; Lippert et al., 2011; Huang et al., 2013, 2015a,b) have proposed several hypotheses to explain the large degree of scatter within the available data, such as compaction-induced inclination shallowing, non-dipolar geomagnetic field geometry, tectonic shortening, or a poorly constrained APWP for East Asia and/or non-rigidity of the Eurasian plate.

The APWP for Eurasia proposed by Besse and Courtillot (2002) is usually employed as a reference curve for Late Mesozoic to Cenozoic paleomagnetic studies in Asia. The Late Cretaceous and Paleogene reference poles for Eurasia are based mainly on data from northwestern Europe, with a few data from eastern Asia and Siberia. On the other hand, recent geological and geophysical evidence suggests that the European platform has not behaved as a stable plate since the Late Cretaceous (Yegorova and Starostenko, 1999; Trifonov, 2004; Cogné, 2013). Based on a compilation of 533 Cretaceous to present-day paleomagnetic poles obtained from both sedimentary and igneous rocks, Cogné et al. (2013) constructed a new APWP for East Asia. Their analysis shows convincingly that the Apparent Polar Wandering Path (APWP) for Europe can no longer be used to interpret paleomagnetic data from east of the Urals, including interpretation of Asian Tertiary deformation related to the India–Asia Collision. Thus, we use the APWP for

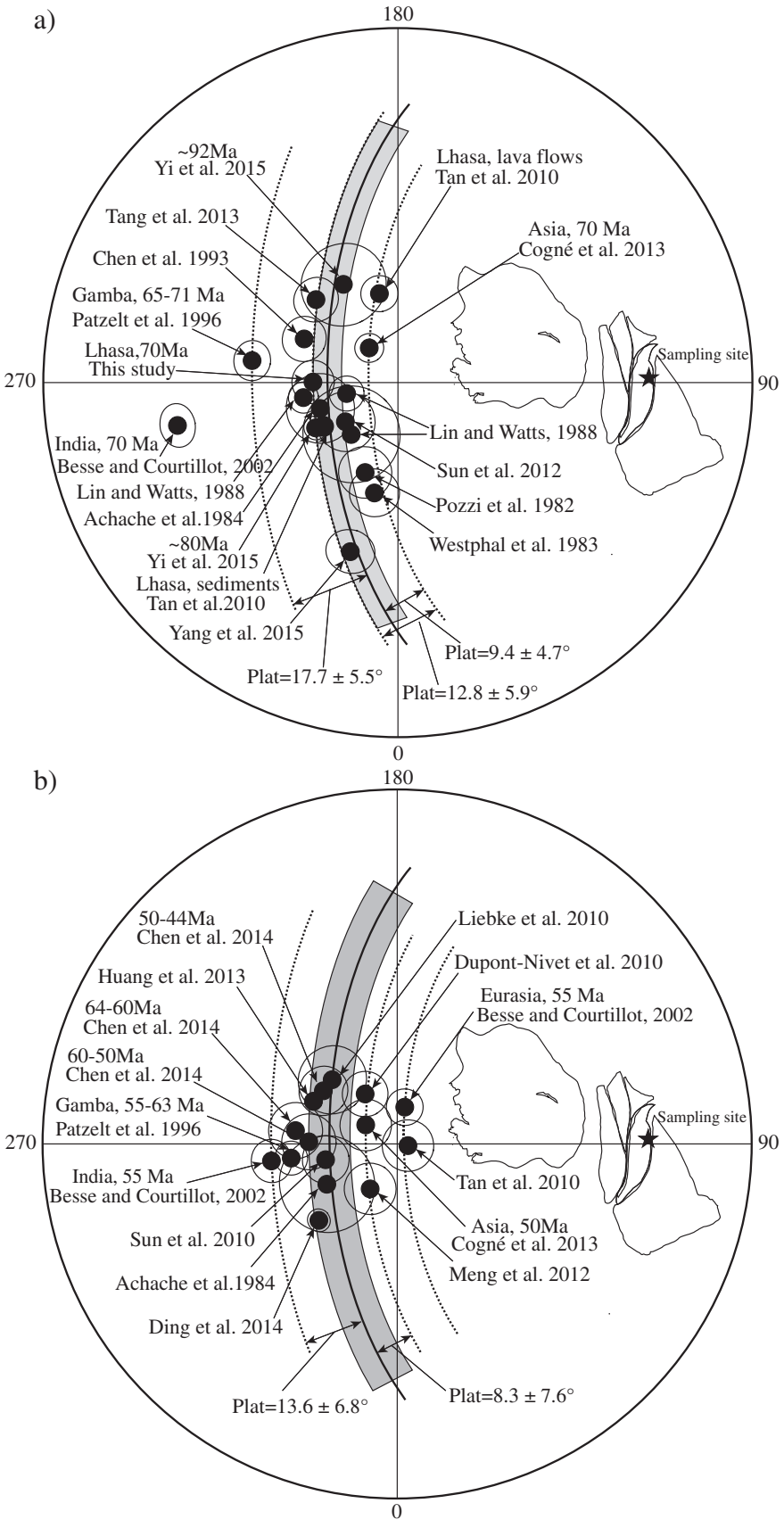


Fig. 8. Equal-area projection of late Cretaceous poles (a) and Paleogene (b) of the Lhasa terrane and adjacent blocks. See Table 3 for values and references. plat = latitudinal displacement; light gray area with bold black line in (a) and (b): small-circle with its 95% confidence area passing through paleopoles, and centered on the average site locations; the black star indicates the sampling location, which was selected as the reference site (29.9°N, 91.1°E) when calculating the paleolatitude of the Lhasa terrane in this study.

Asia proposed by Cogné et al. (2013) to calculate the expected paleolatitude of the Lhasa terrane. The Late Cretaceous and Paleogene paleolatitudes for the southern margin of Asia calculated from the apparent polar wander path are 23.2 ± 3.3°N (70 Ma) and 22.7 ± 5.9°N (50 Ma), respectively.

Our new paleomagnetic results from Late Cretaceous Shexing red beds and interbedded lava flows from the Linzhou basin demonstrate the robustness of the Late Cretaceous paleopole for the Lhasa Terrane based on positive fold and reversal tests, adequate sampling of the Earth's paleosecular variation (PSV) and similar inclinations recorded in both volcanic and sedimentary rocks. These findings indicate that at least some of the Cretaceous sediments in the Lhasa Terrane have experienced no significant compaction-induced inclination shallowing. The paleomagnetic results indicate that the Lhasa Terrane was located at 10.4 ± 4.9°N during the interval of 68–75 Ma. In addition, the combined previous paleomagnetic data indicates that the Lhasa terrane was located at 14.4 ± 4.8°N during Early Paleogene. Our calculations indicate a post-Cretaceous convergence of 12.8 ± 5.9° (1420 ± 650 km) between the southern parts of Tibet and the northernmost parts of Siberia. Based on previous paleomagnetic data, the post-Cretaceous convergence within Asia is about 9.4 ± 4.7° (1040 ± 520 km) (Table 3, Fig. 8a). The total convergence is 8.3 ± 7.6° (920 ± 840 km) between Asia and the Lhasa Terrane, and 13.6 ± 6.8° (1510 ± 760 km) between India and the Lhasa Terrane since 50 Ma (Table 3, Fig. 8b). Within the uncertainties of paleomagnetic methods, this result is in agreement with the geological record of upper crustal shortening in Asia. However, the estimates of the amount of convergence are slightly lower than the previously published estimates of 1700–2000 km of post-Cretaceous intracontinental shortening (Achache et al., 1984; Lin and Watts, 1988; Patzelt et al., 1996; Liebke et al., 2010; Sun et al., 2010, 2012) between southern Tibet and Siberia.

6. Conclusions

We have reported new paleomagnetic results from 21 sites from Late Cretaceous red bed sequences intercalated with volcanic rocks in the Linzhou Basin in the Lhasa terrane. A high temperature characteristic magnetic component was isolated from 10 red bed and 11 lava flow sites. The samples from the red beds passed the fold test and those from the volcanic rocks passed the reversal test, and therefore we argue that the magnetization is primary and is dated to ~70 Ma using the K–Ar method. The consistent inclinations recorded in the red beds and basalt lava flows from the Lhasa terrane indicate the absence of significant inclination shallowing caused by deposition compaction in red beds from the studied section. Based on the presently available results from the Lhasa terrane, the southern margin of Asia was located at a low latitude and maintained a stable paleolatitudinal position (10–15°N) during the Late Cretaceous to the Paleogene. This indicates that the southern margin of the Lhasa terrane was rather straight prior to the India–Asia collision. When the results are compared with the expected paleomagnetic direction from the APWP for eastern Asia of Cogné et al. (2013), the latitudinal difference suggests that the crustal shortening occurring between the southern margin of Asia and the stable part of Asia is 1040 ± 520 km (920 ± 840 km) since the Late Cretaceous (50 Ma).

Acknowledgments

We particularly wish to thank two anonymous reviewers and Zeming Zhang and M. Santosh for their insightful comments and valuable suggestions. This research was funded by NSFC (40874030), by China Geological Survey (1212011121264) and by GPMR (201505).

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